

Donghee Hong

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Research Interests

Visual analytics, human-computer interaction, user-centered design, interactive visualization, human-AI collaboration

Education

Sungkyunkwan University (SKKU)

M.S. IN ARTIFICIAL INTELLIGENCE (EXPECTED FEBRUARY 2026)

- Cumulative GPA: 4.5/4.5

Suwon, Korea

Mar 2024 - Present

Sungkyunkwan University

B.S. IN COMPUTER SCIENCE AND ENGINEERING

- Cumulative GPA: 4.35/4.5 (98.5%), Major GPA: 4.38/4.5
- Graduated Summa Cum Laude, Class Rank: 2/160

Suwon, Korea

Mar 2020 - Feb 2024

Research Experience

Interactive Data Computing Laboratory (IDCLab), SKKU

RESEARCH ASSISTANT (ADVISER: PROF. JAEMIN JO)

- Led, as first author, the development of **HyPockeTuner**, a mobile interactive HPO system featuring EventCrumb, a novel event sequence visualization tailored for smartphones that unified HPO progress, user interventions, and outcomes on a single timeline
- Led research as first author on **Symetra**, a visualization system for analyzing parameters and their impact on symbolic execution engines, and collaborated with domain experts in software testing
- Conducted multiple user studies, including formative studies (N=8), case studies (N=2), deployment studies (N=2), expert interviews (N=2), and controlled lab studies (N=12) with domain experts to evaluate system designs and inform iterative development
- **SAE Feature Visualization** (Sep 2025–Present): Contributed to the development of a web-based interactive dashboard and visualizations for the analysis of Sparse Autoencoder (SAE) features in a neural network interpretability project

Suwon, Korea

Aug 2022 - Present

Data-driven AI Security HCI (DASH) Laboratory, SKKU

RESEARCH ASSISTANT (ADVISER: PROF. SIMON SUNG-IL WOO)

- Participated in various real-world application projects and contributed to the construction of large-scale datasets
- Notably, contributed to the development of a model for detecting fake faces, regardless of mask usage

Suwon, Korea

Dec 2020 - Jun 2022

Publications

UNDER REVIEW

- [6] **Hong, D.**, Kim, M., Cha, S., & Jo, J. (Under review at EuroVis 2026). Symetra: Visual Analytics for the Parameter Tuning Process of Symbolic Execution Engines.
- [5] **Hong, D.**, Lee, B., Seo, J., & Jo, J. (Revise & Resubmit at CHI 2026). HyPockeTuner: Bringing Hyperparameter Optimization to Mobile Devices.

PUBLISHED

- [4] **Hong, D.**, Kim, M., Cha, S., & Jo, J. (2025). Symetra: Visual Analytics for Tuning Symbolic Execution Engines. *IEEE VIS (Poster)*.
- [3] Park, S., Park, J., & **Hong, D.** (2024). LLM-Based Text-to-Image Prompt Conversion System for Custom Wheelchair Spoke Guard Design. *Proceedings of the Korean Computer Science Society Conference*, 918–920.
- [2] Lee, S., Ko, D., Park, J., Shin, S., **Hong, D.**, & Woo, S. S. (2022, May). Deepfake Detection for Fake Images with Facemasks. *Proceedings of the 1st Workshop on Security Implications of Deepfakes and Cheapfakes*, 27–30.
- [1] An, J., Kim, J., Lee, H., Kim, J., Kang, J., Shin, S., Kim, M., **Hong, D.**, & Woo, S. S. (2021, December). VFP290k: A Large-Scale Benchmark Dataset for Vision-Based Fallen Person Detection. *The Thirty-Fifth Conference on Neural Information Processing Systems Datasets and Benchmarks Track (Round 2)*.

Presentation

Hong, D. (2024, Oct). ViSTrics: A visual analytics tool for software testing. Korean Visualization Workshop (K-VIS), Seoul, Korea.

Selected Projects

PediaViz: Interactive Timeline Dashboard for Pediatric Medication and Test Analysis

IDCLAB, SKKU COLLABORATING WITH SEOUL NATIONAL UNIVERSITY HOSPITAL & NAVER CORPORATION

Feb 2024 - Present

- Integrating natural language diagnostic data into the system, creating a multimodal system that seamlessly combines structured clinical measurements with unstructured textual assessments for comprehensive patient analysis
- Designed and implemented a web-based interactive dashboard that synchronizes patient growth records, lab results, and medication history on a unified timeline
- Built custom visualization components to help pediatricians identify patterns and correlations between treatments and test results

Spokeguard Design Assistant for Young Wheelchair Users

CO-DEEP LEARNING PROJECT, SKKU

Sep 2023 - Dec 2023

- Developed an AI model that converts Korean design descriptions into appropriate image-generation prompts for young wheelchair users
- Addressed the accessibility gap for children with disabilities by eliminating the need for complex English prompts in text-to-image platforms

License Plate Recognition Mobile Application

UNDERGRADUATE RESEARCH PROGRAM (URP), SKKU

Jan 2021 - Sep 2021

- Built end-to-end pipeline from dataset construction and model training to mobile application deployment
- Published on Google Play Store with 1,000+ downloads

Teaching Experience

TEACHING ASSISTANT

Introduction to Human-Computer Interaction (Fall 2024, Fall 2025); Information Visualization (Spring 2025); Problem Solving (Fall 2025); Computer Programming for Engineers (Spring 2023, Spring 2025); Python Programming Orientation for Freshmen (2021-2023)

TUTORING AND MENTORING

SKKU Peer Tutoring Program (Spring 2021); SKKU Software Mentoring Program (2020-2021)

Awards and Honors

SCHOLARSHIPS

Graduate Scholarship, SBS CULTURAL FOUNDATION

Sep 2024 - Feb 2026

- Full tuition and annual stipend of \$8,225
- Around 15 students selected nationwide (acceptance rate: 4%)

Graduate Scholarship, DAESANG FOUNDATION

Mar 2024 - Jun 2024

- Stipend of \$4,800 per semester

The Presidential Science Scholarship, KOREA STUDENT AID FOUNDATION

Mar 2022 - Dec 2023

- Full tuition and annual stipend of \$3,500
- Approximately 150 students selected nationwide

SKKU Software Scholarship, SKKU

Mar 2020 - Dec 2022

- Full tuition for two years (contingent on maintaining an overall GPA above 3.5)

Additional Internal Scholarships, SKKU

- Extracurricular Challenge Scholarship (2023); Student Success-Leadership Scholarship (2021, 2022); Department Recommendation Scholarship (Fall 2021)

AWARDS

Honorable Mention, Graduate Student Paper Award , SKKU	Dec 2025
Grand Prize, Co-Deep Learning Project Competition , SKKU Ranked top 2 out of 16 teams	Dec 2023
Honorable Mention, SW-Centered College AI Competition , SW-CENTERED UNIVERSITY ASSOCIATION Ranked top 8 out of 194 teams	Dec 2023
Dean's List Award, College of Computing and Informatics , SKKU	Spring & Fall 2020, Fall 2021

Leadership and Extracurricular Activities

SKKU-BA-DIVE Program TEAM LEADER Led field research and interviews with industry professionals in IT and AI	London, UK Jan 2023
Uni-DTHON Competition DATA-THON TRACK LEADER Coordinated AI competition planning across six major universities in South Korea, designing competition problems and implementing leaderboard evaluation	Seoul, Korea Dec 2021 - Dec 2022
Student Council, College of Computing and Informatics VICE PRESIDENT Organized in-person academic events and facilitated communication between students, faculty, and partner universities	Suwon, Korea Nov 2021 - Nov 2022
University Student Council Joint Committee COLLEGE REPRESENTATIVE Represented the College of Computing and Informatics to discuss and decide on university development issues	Seoul, Korea Nov 2021 - Nov 2022
Global Challenge Program DEPUTY TEAM LEADER Conducted field research and interviews with IT professionals	Silicon Valley, USA Jun 2022
Community Service VOLUNTEER Over 100 hours of community service	Suwon, Korea 2020 -2024

Technical Skills

Research Methods	Semi-structured interviews, Case Studies, Controlled Lab Studies, Formative Studies, Thematic Analysis
Programming	TypeScript and JavaScript (Advanced, 3+ years production experience), Python (Advanced), Java, C and C++
Visualization & Frontend	D3.js & visx (Advanced, custom interactive visualizations), React (Advanced), Tailwind CSS, HTML/CSS
Data Analysis & Backend	Pandas, PyTorch, NumPy, Scikit-learn, FastAPI
Tools	Git, Figma, LaTeX, Adobe Premiere Pro